Smart Streetlight Model: Sensor Selection & Architecture Matrix

1. AI SENSOR-FUSION COMPONENT SELECTION

Sensor / Module	Primary Role	Target Detected	Vehicle Logic Behavior	Pedestrian Logic Behavior	Recommended Model / Part No.	Hardware Interface
Microwave Doppler Radar	Long-range velocity and approach tracking.	Vehicles & Fast Moving Objects	Triggers 100% optimum brightness instantly when high-frequency shift (>15 km/h) is tracked ahead.	Ignored if frequency shift is low and target matches pedestrian signature, or used as a secondary wake-up.	HB100 / CDM324 (or RCWL-0516 for simple binary models)	Analog Input / Frequency Counter Pin
Passive Infrared (PIR) Array	Short-range thermal footprint confirmation.	Humans / Pedestrians (Heat signatures)	Ignored. Car chassis blocks direct body heat; logic treats it as non-human based on zero PIR activity.	Triggers a localized 'Hold Phase' at 40-50% brightness to ensure walking safety along sidewalks.	AM312 Mini PIR / Panasonic PaPIRs	Digital GPIO (Interrupt Pin)
Edge Camera Module	Computer Vision object classification & size boundary checking.	Both (Classifies 'car', 'truck', 'person')	Runs lightweight YOLOv8-nano/MobileNet. Determines distance via bounding box height; scales light up dynamically.	Classifies 'person'. Locks brightness to comfortable baseline; dims immediately once bounding box disappears.	ESP32-CAM / Omnivision OV2640 / Raspberry Pi Cam v2	CSI / SPI / Dedicated Camera Interface
Digital Ambient Light Sensor (ALS)	Measures ambient environmental lux levels.	Natural Environment (Cloud, Dusk, Solar)	Prevents false dimming. System ignores passing headlights because Master Clock overrides daytime behaviors.	Ensures baseline safety light floors are raised during heavy storms or unexpected low-visibility atmospheric events.	BH1750 / TSL2591	I2C Protocol (SDA/SCL Pins)
Real-Time Clock (RTC) Module	Chronological time scheduling keeper.	Time Gates (Curfews / Active Hours)	Enforces strict midnight curfews. Post-midnight, caps vehicle max brightness at 50% to mitigate extreme skyglow.	Maintains lower safe pedestrian illumination scales during designated low-traffic late-night hours.	DS3231 (High Precision with TCXO)	I2C Protocol (SDA/SCL Pins)

2. DISTANCE-BASED AUTOMATION AND DIMMING LOGIC RULESET

Target Class	Detection Distance / Window	Target LED PWM Output	System Adaptive Transition Behavior	Light Pollution Prevention Mechanism	AI Inference Processing	Sensor Fusion Dependencies
Vehicle	Approach Zone (30 - 40 meters out)	100% Brightness (PWM: 255)	Instantly ramps up up-ahead lamps using a networked cascade loop so the driver always looks into illuminated space.	Limits full-beam operations strictly to when a vehicle is traveling through the sector.	YOLOv8 bounding box height is small but expanding; Radar senses positive Doppler shift.	Microwave Radar + Edge Camera
Vehicle	Departure Zone (Passing Away)	Linear Decay to 10% Baseline	Slowly fades down over a 3-5 second decay window. Prevents sudden flashing/strobing behind the rearview mirror.	Eliminates late-night residual skyglow on empty thoroughfares once passing safety distance is satisfied.	Bounding box diminishes; Radar changes to a negative Doppler shift frequency pattern.	Microwave Radar + Edge Camera
Pedestrian	Sidewalk Proximity (0 - 10 meters)	40% - 50% Brightness (PWM: 115)	Maintains a warm, comfortable safety envelope. Light stays on continuously even if the human stops moving.	Saves more than 50% energy compared to vehicle illumination limits; narrows light strictly to human paths.	AI matches structural 'person' class features; PIR confirms continuous localized body heat spikes.	PIR Sensor + Edge Camera
False Trigger	Any distance (e.g. Swaying tree/animal)	No Change (Stays at 10% Eco-Mode)	The system completely filters out the movement. No lighting change is introduced.	Prevents unnecessary ecosystem and wildlife disruption caused by typical hardware sensor glitches.	TinyML network classifies the input wave spectrum as a background disturbance anomaly.	Radar Data Pattern + Edge AI Model